



Part I

Final Report

FitML: A Fairness-Audited Machine Learning System for Sizing and Intelligent
Virtual Try-On in E-Commerce

FitML: A Fairness-Audited Machine Learning System for Sizing and Intelligent Virtual Try-On in E-Commerce

Pillar 5: Capstone Project

Postgraduate Diploma in Artificial Intelligence and Machine Learning

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Live demo: <https://fit-ml.netlify.app>
Repository: <https://github.com/debiegalleros/fitml-capstone>

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1 Problem Framing

Business problem. Online clothing returns run at roughly 35%, versus about 8% in physical stores, costing retailers over \$100B annually. Sizing uncertainty is the leading driver, but shoppers also hesitate because they cannot see how a garment will actually look on their own body. FitML addresses both halves of that hesitation: a fairness-audited sizing model answers “will it fit?”, and a generative virtual try-on composites the actual catalog garment onto the shopper’s own photo before buying — answering “how will this look on me?”. Color-suitability matching (does this shade complement the shopper) was identified as a natural extension and deliberately scoped out of this submission.

Data science problem. Supervised multi-class classification: given a customer’s body measurements and the size they ordered, predict the reported fit outcome — fit, runs small, or runs large — paired with a fairness-audit layer that checks whether predictions are equally reliable across body types, heights, and size ranges. The labels come from real customer fit feedback, not collaborative filtering.

Success metrics. Technical: accuracy and macro-F1 (macro-F1 catches the minority-class failures that raw accuracy hides under a 73%-majority class), plus disparate impact ratio and equalized odds for the fairness layer. Business: projected return-rate reduction (35% toward a 15–20% target), with fewer wrong “it fits” reassurances as the mechanism.

The differentiator. Commercial sizing tools are black boxes. FitML publishes its full bias audit — per-group metrics, disparate impact, equalized odds, SHAP explainability, and an honest before/after mitigation comparison — as a first-class deliverable alongside the product.

2 Data Collection & Understanding

FitML deliberately combines sources with different, clearly-flagged statuses — real audited data where it exists, and a small synthetic extension where it does not:

Need	Source	Status
Model training + full fairness audit (graded core)	Kaggle “Clothing Fit Dataset for Size Recommendation” — 251,047 real customer reviews (58,503 ModCloth + 192,544 RentTheRunway) with self-reported measurements and fit feedback	Real
Men’s catalog extension (~11 items, demo only)	200 synthetic customers labeled by rule-based lookup against Uniqlo’s published men’s size charts	Synthetic population, real sizing standard — excluded from the fairness audit
Garment catalog imagery (visual only, never training data)	Kaggle Fashion Product Images Dataset (MIT license) — 117 curated items, background-removed via rembg, 234 hue-shifted color variants	Real photos, demo pricing

Why this combination. Both ModCloth and RentTheRunway sell women’s apparel, so the real dataset is women’s-only. Rather than force one dataset to cover both genders — or silently train on invented men’s fit feedback — the project uses real, citable data where it is available and a small, explicitly-documented synthetic extension for the gap. The men’s extension is a deterministic size-chart lookup, not a trained model, and its ~200-row synthetic population is far too small for valid group comparisons — so it is excluded from the fairness audit by design, an explicit documented limitation rather than a silently patched one.

Key fields (full data dictionary in docs/data_dictionary.md). Height, weight, bust band + cup, hip, self-reported body type (hourglass / pear / apple / athletic / petite / straight & narrow / full bust), garment category, ordered size, and the target fit_feedback (small / fit / large). Waist was dropped: RentTheRunway never collected it and ModCloth's coverage was 3.5% (~96.5% missing), leaving <1% combined coverage — too sparse to impute defensibly. Post-purchase fields (rating, review text) were excluded as features because they would not exist at prediction time.

Catalog imagery provenance. iMaterialist Fashion was evaluated first and rejected: its images are in-the-wild street-style, red-carpet, and runway photos — busy backgrounds, identifiable people, some with third-party photographer watermarks — a poor fit both technically (for background-removal + affine-warp compositing) and ethically. The Fashion Product Images Dataset provides clean, plain-background product shots with structured gender/category/color metadata; only the ~115 needed full-resolution images were fetched via targeted single-file downloads, not the full 24.7 GB archive. Catalog prices are programmatically generated demo pricing within per-category PHP bands, documented as such.

3 Preprocessing, EDA & Feature Engineering

Full walkthrough with reasoning in notebooks/02_eda_feature_engineering.ipynb; the reusable logic lives in src/prepare_data.py so later phases import it instead of duplicating it. Output: data/processed/model_ready.csv — 251,047 rows × 15 columns, no remaining nulls.

Harmonizing two schemas

The sources disagree on units and formats (height as “5ft 6in” vs “5' 8””, hips in inches, bust as band+cup strings). Parsing normalizes everything to metric. ModCloth rows tagged only new/sale/wedding (24,287 rows, ~29% of ModCloth) carry no garment-category signal and were dropped; RentTheRunway's 68 fine-grained category tags were mapped onto ModCloth's 4-bucket taxonomy (tops / dresses / bottoms / outerwear) as category_broad. Cup letters were unified to an ordinal scale (AA=0 ... K=11) using the standard DD=E, DDD=F, DDDD=G equivalence.

Structural missingness, handled honestly

Missingness here is structural, not random: weight is absent for all ModCloth rows (never collected), hips for all RentTheRunway rows, body type for all ModCloth rows. Instead of silently imputing and pretending the values are measured, each such field gets a _missing indicator flag alongside median imputation, so models — and the fairness audit — can distinguish a measured value from an invented placeholder.

body_type gets an explicit not_reported category rather than a guessed shape: inventing a protected-attribute-like label would contaminate the Phase 6 audit. This decision pays off later — SHAP shows the model actually uses these missingness flags (Section 5.3).

Class imbalance in the target

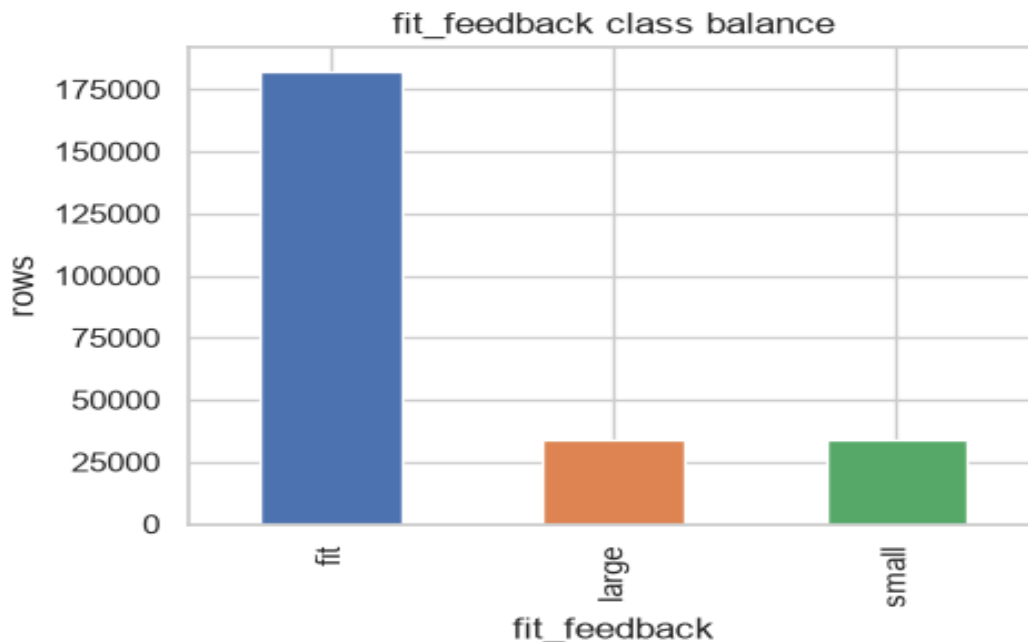


Figure 1. Target distribution: 72.7% fit, 13.7% large, 13.6% small. Severe class imbalance (73%/13%/13%) necessitates macro-F1 as the selection metric; raw accuracy is dominated by the majority class.

The target is ~73/14/13 — exactly why macro-F1, not accuracy, drives model selection: a model predicting “fit” always achieves ~72.7% accuracy while failing entirely on the minority classes that need size corrections.

Distributions and correlations

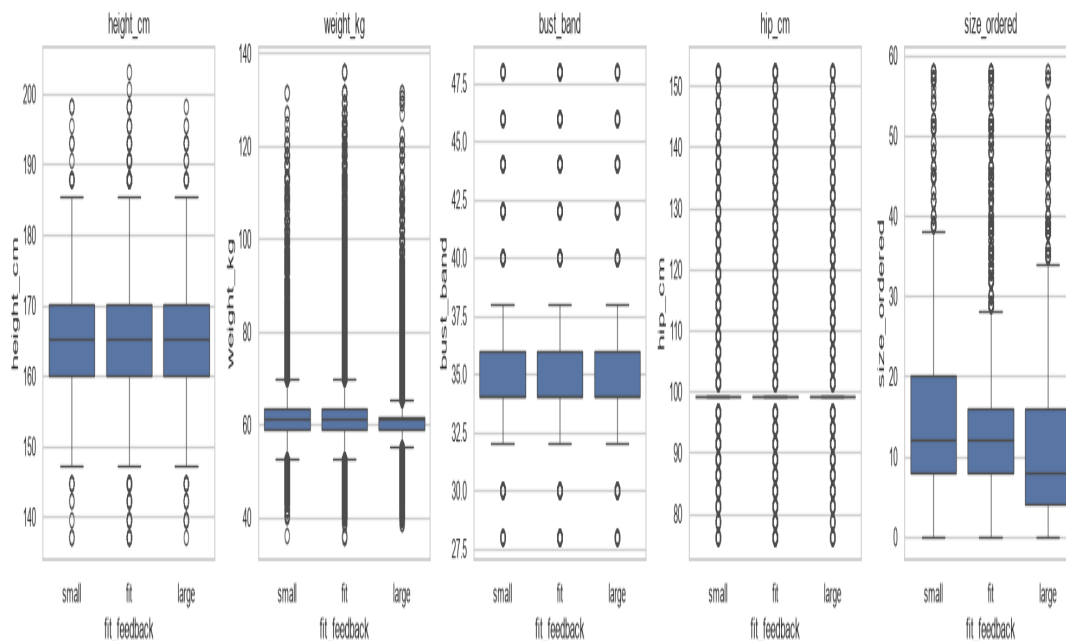


Figure 2. Feature distributions per fit_feedback class. size_ordered separates most clearly; body measurements separate weakly on their own.

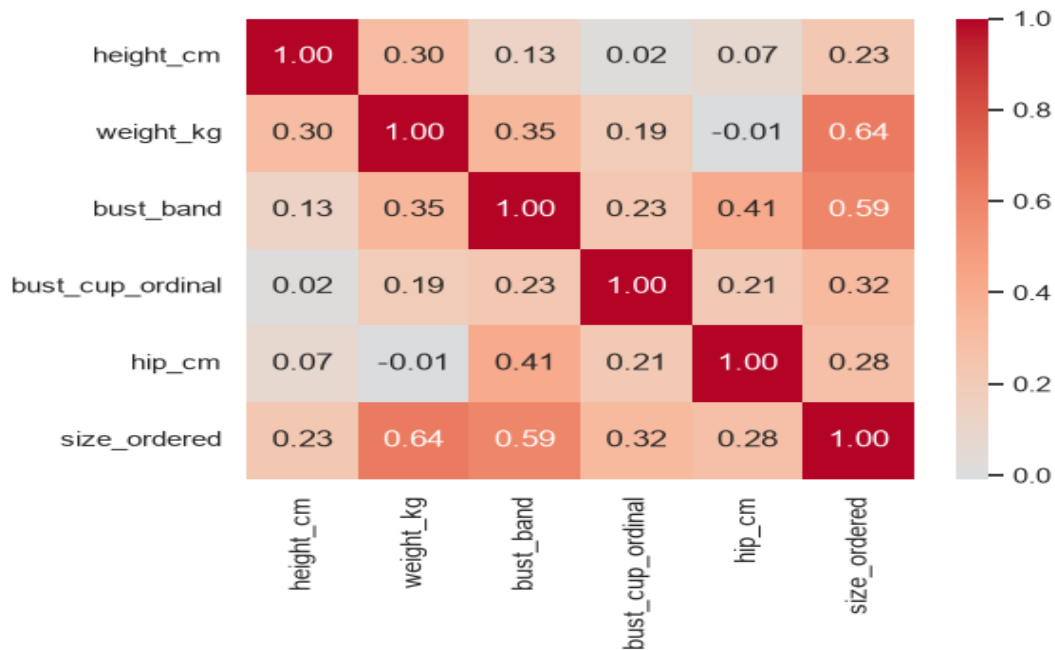


Figure 3. Correlation heatmap of numeric features.

size_ordered shows the clearest class separation (it is mechanically tied to fit), while body measurements separate weakly in isolation — fit depends on the relationship between measurements and the ordered size, not on measurements alone.

PCA and exploratory feature importance

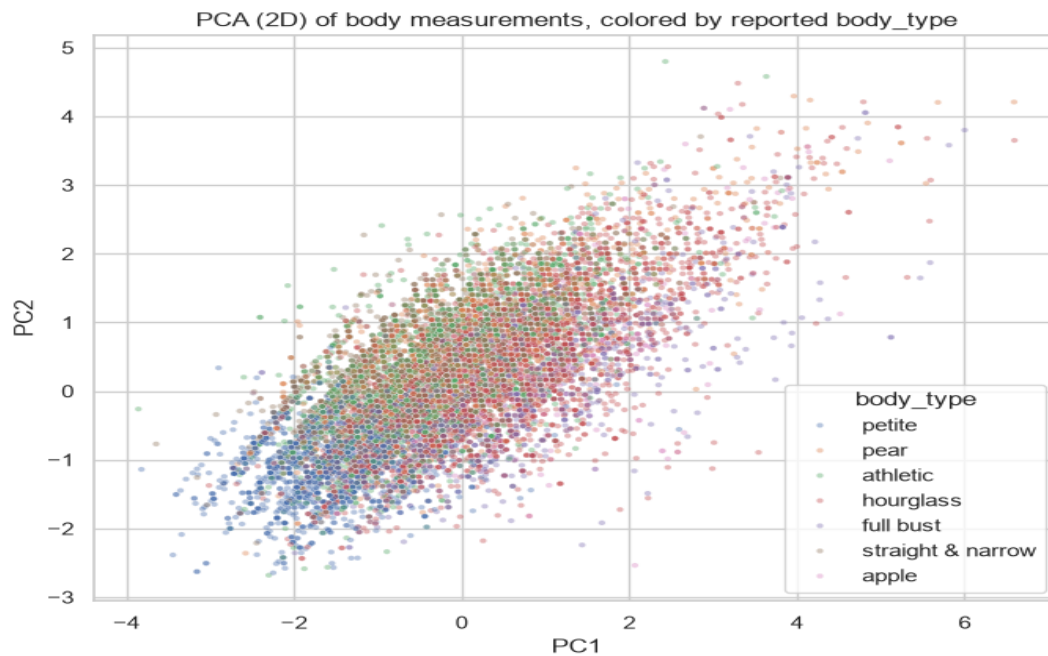


Figure 4. PCA (2D, visualization only) colored by body type.

Body-type clusters overlap substantially in 2D: PC1/PC2 are dominated by overall body size, while body-shape categories are defined by proportional relationships (e.g. waist-to-hip ratio). This is a useful negative result — raw measurements will not trivially separate body types, reinforcing why the fairness audit needs per-group metrics rather than assuming a simple boundary.

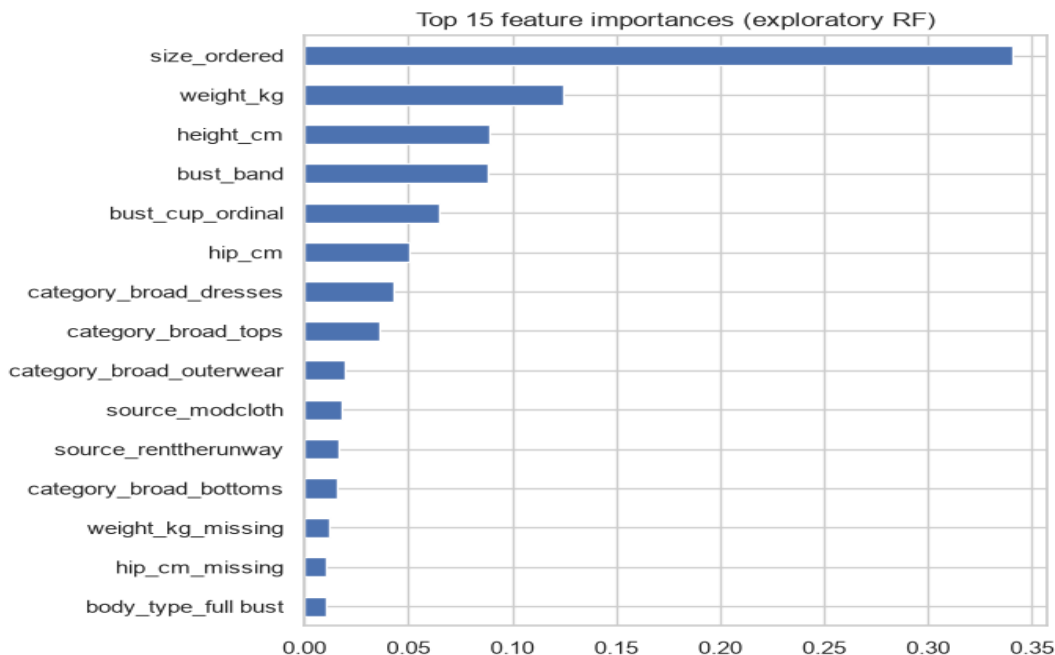


Figure 5. Exploratory random-forest feature importances (fixed seed).

Leakage discipline. `model_ready.csv` ships deliberately unscaled and not one-hot encoded: scaling and encoding are fitted inside each Phase 5 model's train-only sklearn Pipeline, so no test-split statistics ever leak into training.

4 Model Implementation & Comparison

Four classifiers — Logistic Regression (baseline), Random Forest, XGBoost, and a two-hidden-layer MLP (64/32, early stopping) — were trained by `src/train_models.py` on the same stratified 80/20 split (200,837 train / 50,210 test rows, seed 42), same features, same untuned equal-footing hyperparameters, with preprocessing fitted inside each pipeline on the training fold only.

model	accuracy	macro-F1	F1 fit	F1 large	F1 small	train time (s)
xgboost	0.7272	0.2918	0.8418	0.0219	0.0115	2.7
random_forest	0.7278	0.2848	0.8424	0.0101	0.0018	3.4
mlp	0.7272	0.2825	0.842	0.0043	0.0012	6.1
logistic_regression	0.7267	0.2822	0.8418	0.0	0.0046	0.8

Table 1. Phase 5 comparison (`models/comparison.csv`). Selected: XGBoost, by macro-F1.

Why macro-F1

Accuracy is statistically indistinguishable across all four models — each lands at the 72.7% majority baseline — which is precisely why it was ruled out as the selection metric. Macro-F1 weights each class equally, exposing minority-class failure: the “small” and “large” predictions are the entire point of a sizing assistant, because they are the cases where a recommendation prevents a return. XGBoost wins as the only model with non-trivial F1 on both minority classes.

The honest headline: unweighted models collapse to the majority class

Every unweighted model learns to say “fit” almost always (logistic regression never predicts “large” at all). Two causes: (1) with 73% of labels being “fit”, an unweighted loss is minimized by rarely risking a minority prediction; (2) the profile features carry real but limited signal — fit feedback also depends on garment-specific cut and sizing quirks no body measurement captures, and published work on this same dataset reports the task as hard for the same reason. This was deliberate for Phase 5: the unweighted models are the honest “before” for the fairness mitigation in Section 5.

Trees vs. the MLP — the expected finding, reported honestly

The Master Plan predicted tree-based models would beat the MLP on this modest-size, mixed-type tabular dataset, and they did: the MLP lands below both XGBoost and Random Forest on macro-F1 despite the longest training time. This matches the well-documented pattern that gradient-boosted trees dominate MLPs on tabular data of this scale — a legitimate comparison result, not a tuning failure. CNNs and diffusion models were explicitly scoped out.

Which model the product serves

The deployed /recommend-size endpoint serves the class-weighted XGBoost from the Section 5 mitigation, not the unweighted Phase 5 winner. For a sizing assistant the costly error is a missed misfit — the unweighted model gives wrong “it fits” reassurance in ~96–100% of misfit cases, while the weighted model catches roughly half of misfits (recall large 0.486, small 0.546). The weighted model is also better on the selection metric itself (macro-F1 0.3596 vs 0.2918). Its accuracy cost (0.727 → 0.396) is surfaced, not eliminated, by the product layer: the endpoint returns class probabilities, and the displayed confidence % plus a blue/amber advisory box communicate uncertainty, so borderline predictions surface as a sizing tip, never a bare verdict.

5 Critical Thinking, Ethical AI & Bias Audit

Full report in docs/fairness_report.md; every number below is generated by src/fairness_audit.py, which re-derives the exact Phase 5 split (seed 42) and asserts it matches the pinned test indices before computing anything. The audit covers the real women's model only — the men's synthetic extension is excluded because group metrics on a ~200-row generated population would measure properties of the random-number generator, not real-world disparity.

5.1 Setup: harms, groups, metrics

A “fit” prediction is the favorable outcome (a confident go-ahead at the usual size); small/large predictions route the shopper into size-adjustment friction. Two harms matter: quality-of-service harm (misfit detection failing more for some groups → more wrong reassurance → more returns) and allocation-style harm (the favorable prediction handed out at unequal rates). Groups audited, each with $\geq 1,000$ test rows: body_type (8 groups; “not_reported” is the plurality at $n=14,800$ of 50,210 test rows, 29.5%), height_band (under 160 / 160–169 / 170+ cm), and size_band (US 0–6 / 8–14 / 16–22 / 24+). Metrics: per-group accuracy and per-class recall, disparate impact ratio

(four-fifths rule, <0.8 flagged), and equalized-odds TPR/FPR spreads.

Audit scope. Groups audited here are morphological proxies, not demographic protected attributes: the dataset contains no age, race, disability, or gender fields, and the population is women-only (both ModCloth and RentTheRunway serve women's fashion). `body_type`, `height_band`, and `size_band` are the fit-relevant dimensions this clothing-recommendation domain actually supports, and the audit stratifies on all three. Results measure disparity in model performance across these groups; they are not a substitute for an audit of demographic protected attributes, which the data does not contain.

5.2 Baseline audit: “fair” numbers that mean nothing

No group is DI-flagged at baseline (all ratios ≥ 0.98) — and that apparent fairness is vacuous. The unweighted model predicts “fit” for 99.5% of all rows in every group; a model that always says yes passes the four-fifths rule by construction. The metrics that look inside the predictions show real disparities:

- Accuracy tracks body size: size 0–6 shoppers get 0.7506, size 24+ get 0.6781 — a 7.3-point gap against plus-size shoppers, who experience misfit more often and get the least reliable model exactly there.
- Misfit detection is near zero for everyone, but unevenly so: full-bust shoppers get their “runs large” cases caught 8.2% of the time, pear-shaped shoppers 0.34% — a 24× ratio. The “small” class has literally zero recall for apple body types and size 0–6.
- The fit-class false-positive rate is 0.96–1.00 in every group: when an item does not fit, the model tells virtually every shopper that it does.

Conclusion: the baseline's bias problem is not unequal favorable-outcome rates; it is a degenerate predictor whose failures concentrate on the shoppers most likely to experience misfit. The mitigation must attack minority-class recall.

5.3 Explainability: SHAP on the winning model

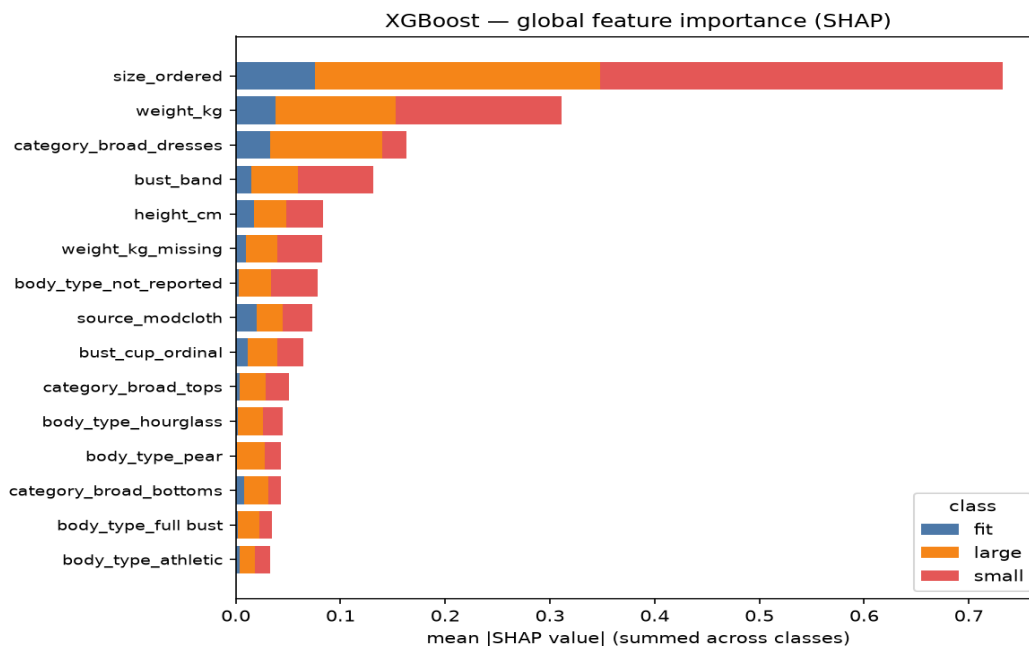


Figure 6. Global SHAP feature importance (TreeExplainer, 3,000 test rows, seed 42).

size_ordered dominates, especially for the minority classes (mean |SHAP| 0.38 for small, 0.27 for large vs 0.08 for fit), and the beeswarms read sensibly: ordering a high size, low weight, and a small bust band all push toward predicting “runs small”. The model has learned real, directionally correct sizing signal — it is the decision threshold, drowned by class imbalance, that fails.

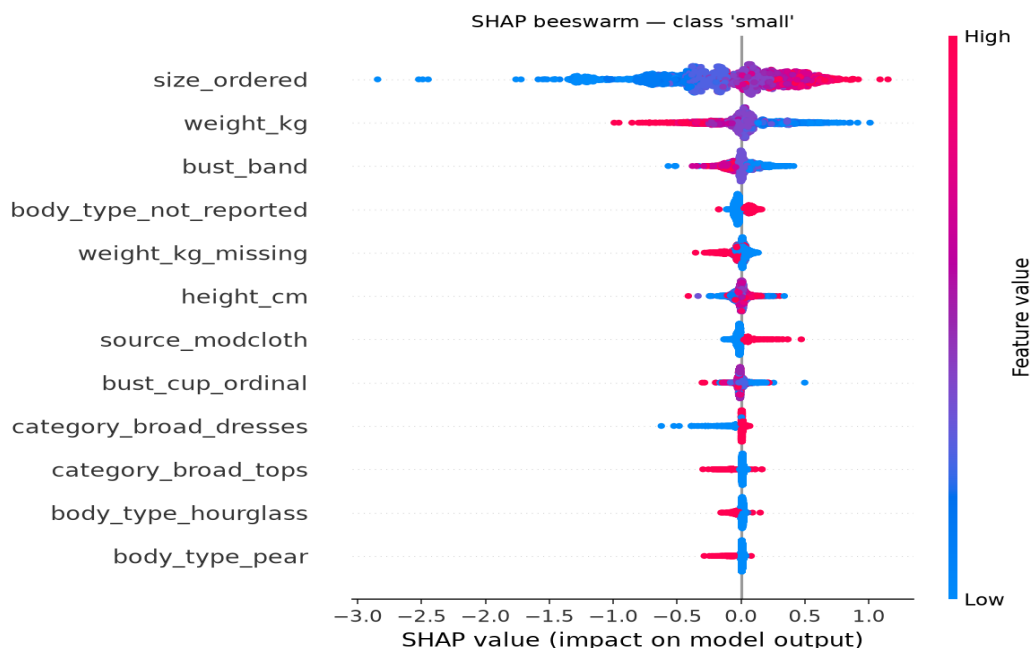


Figure 7. SHAP beeswarm for the “small” class.

Fairness-relevant red flag: missingness is predictive. weight_kg_missing and body_type_not_reported rank in the top seven features for minority classes. These overlap heavily with source_modcloth (weight is missing for all ModCloth rows): the model partly keys on which platform the shopper came from and how much they disclosed, not on their body. Shoppers who decline to report body type or weight get systematically different predictions — a data-completeness bias the product answers with transparency (the confidence box lowers displayed confidence when key measurements are absent).

5.4 Mitigation: class-weighted retraining, before/after

One mitigation, fully audited, as scoped: identical pipeline, identical train fold, identical seed — the only change is balanced sample weights ($\text{weight} \propto 1/\text{class frequency}$) passed to XGBoost.

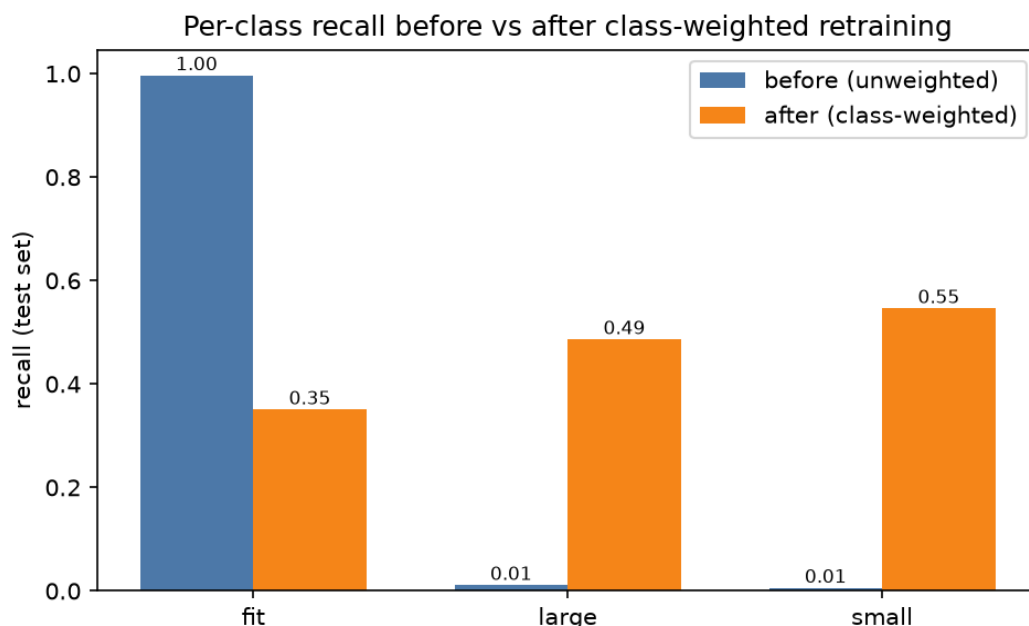


Figure 8. Per-class recall, before (unweighted) vs after (class-weighted).

metric	before (unweighted)	after (class-weighted)
macro-F1 (selection metric)	0.2918	0.3596
recall — large	0.0112	0.4864
recall — small	0.0059	0.5463
min per-group large recall (body_type)	0.0033	0.3553
min per-group small recall (body_type)	0.0000	0.4849
worst-group fit FPR (wrong "it fits")	0.9969	0.3463
accuracy	0.7272	0.3959
DI flags (body_type + size_band)	0 (vacuous)	7 (informative)

Table 2. Mitigation before/after — the fairness-critical numbers.

Reading the trade honestly: misfit detection now exists for every group (the wrong-reassurance rate dropped from ~0.96–1.00 to 0.13–0.35 everywhere), and macro-F1 rose 23%. The costs are equally explicit: accuracy fell to 0.396, most “it fits” cases now receive a spurious caution, and — importantly — disparate-impact flags appear. With the model actually discriminating between classes, the favorable prediction goes to size 8–14 shoppers at rate 0.42 but size 24+ at 0.17 (DI 0.41, flagged), and apple / full-bust / not_reported body types flag at DI 0.64–0.70. Height stays equal (DI ≥ 0.99). These flags are informative, not an artifact: the baseline’s tiny spreads were a side effect of predicting one class everywhere, not evidence of equal treatment. The weighted model partly reflects real, higher misfit base rates in those groups, but at ratios well past the 0.8 threshold and with spurious “runs small” warnings (FPR up to 0.60) for size 16+.

5.4.1 Data-completeness as a fairness axis

Section 5.3 flagged `weight_kg_missing` and `body_type_not_reported` as top-seven SHAP features for the minority classes, overlapping with `source_modcloth` (weight is missing for all ModCloth rows). This raises an obvious question: do the DI flags above reflect real group disparities, or are they a data-collection artifact of which platform a shopper happened to use? A stratified check against the pinned test set answers it directly. Four of the five DI-flagged `body_type` groups — apple, full bust, hourglass, pear — are **100% RentTheRunway rows (0% ModCloth)**, because ModCloth never collected `body_type` at all; only the `not_reported` group is source-mixed (79.4% ModCloth, 20.6% RentTheRunway shoppers who declined to answer). The two DI-flagged `size_band` groups fare the same way: ModCloth's share of size 16–22 (20.7%) and size 24+ (25.5%) sits within a point or two of the unflagged bands (21.5% and 25.1%) and the overall test-set average (23.4%) — flat across the board, not concentrated in the flagged groups. `weight_kg_missing` is also not a pure ModCloth artifact: 15.9% of RentTheRunway rows have it too, and the rate still varies by `body_type` (8.1–21.7%) within the all-RentTheRunway groups. **Conclusion:** the disparate-impact flags on `body_type` and `size_band` are not primarily a platform-missingness confound — they reflect disparities the class-weighted model has genuinely learned within a single, same-source population. The one flag that is source-mixed (`not_reported`) is the one place this audit's missingness caveat still applies at face value. Mitigation (future work): stratified decision thresholds by data-completeness would still be the correct fix for the `not_reported` group specifically; the confidence box's lower displayed confidence when key measurements are absent is a disclosure today, not a calibrated mitigation.

Deployment stance. The product serves the class-weighted model: catching misfits matters more than raw accuracy for a sizing assistant. The confidence layer communicates uncertainty — a blue checkmark for confident matches, an amber “Sizing tip” box instead for borderline ones — so predictions surface as advisory, not bare verdicts. It does not eliminate the underlying false-positive rate: for size 16+ shoppers, up to 60% of “runs small” warnings are wrong (FPR 0.60, Table 2). The amber box contextualizes that false alarm; it does not remove it. A calibrated middle ground — probability calibration plus per-group decision thresholds tuned to equalize misfit-detection rates — is the clear next step, documented as future work to keep the before/after comparison clean.

5.5 Leakage checks and limitations

- Preprocessing fitted train-only inside pipelines; the audit asserts the re-derived split matches the pinned test indices; no post-purchase fields among features; `size_ordered` is legitimately known at prediction time.
- Residual risk, documented: the random split does not group by customer, so a repeat customer could appear on both sides. Customer IDs were dropped upstream; noted as a known limitation.
- Self-reported measurements are noisy, plausibly in group-correlated ways (social-desirability bias in weight reporting is well documented).
- The largest `body_type` group is “declined to say” (n=14,800), so per-group numbers for reported categories are conditioned on willingness to self-categorize.
- Bust input-method (band+cup native vs converted chest measurement) is a fairness-relevant data-quality axis the training data cannot audit; the product logs a per-user `bust_input_method` flag specifically so a future audit can stratify by it.

6 Deployment

The audited model (Section 5) is live end-to-end, not just trained and benchmarked. This section covers the deployment itself (6.1) and the product it serves (6.2).

6.1 Live end-to-end deployment

Live end-to-end on free tiers: the Flask backend runs as a Docker web service on Render (<https://fit-ml.onrender.com>) and the static frontend on Netlify (<https://fit-ml.netlify.app>). Free tiers spin down when idle — the first request after idle can take 30–60 seconds to cold-start.

- Docker was required because MediaPipe's Tasks API loads GPU-stack shared libraries (libGLESv2) even on the CPU delegate; the Dockerfile installs the needed system libraries that Render's native Python runtime cannot.
- Catalog images are gitignored, so a build-time script fetches them from a GitHub Release asset — keeping binary blobs out of git history while making assets available on the ephemeral-disk host.
- The full flow was verified against the live backend: photo upload (with pose extraction and the opt-in privacy crop) → size recommendation → generative try-on → Claude advice call.
- The `ANTHROPIC_API_KEY` lives only in the host's environment variables, never in the repository.
- Scope note: the live demo uses only the fairness-audited women's dataset and the class-weighted XGBoost model. The men's synthetic extension (Section 2, ~200 rows, deterministic size-chart lookup) is included in the repository as a scoping example but is not deployed on the live site; it is excluded from every fairness audit by design.

6.2 The FitML prototype: try-on, advice & privacy

A functional prototype website (not an online store — no checkout, payments, or inventory), live at <https://fit-ml.netlify.app> with six pages: Landing, About/How-it-works, Profile setup, Catalog, Try-on result, and History. Flask backend, plain HTML/CSS/JavaScript frontend (no framework), responsive from 375px mobile to wide desktop.

Generative virtual try-on

MediaPipe extracts pose keypoints from the user's uploaded photo; Claude Vision then analyzes the shopper photo and the target garment image and returns a structured JSON prompt (pose, garment placement, fit cues) that drives three garment-conditioned rendering engines, tried in order per request. **IDM-VTON** (garment-conditioned diffusion, via Replicate) is the default: it takes the shopper's photo and the actual catalog garment cutout directly, with masking and blending happening inside the model. On failure, the pipeline falls back to **SDXL inpainting** — a documented, root-caused limitation of this fallback: the community-maintained cog wrapper does not reliably implement mask-constrained inpainting the way the standard diffusers pipeline does, so it consistently renders a garment resembling the shopper's original clothing rather than the requested catalog item (tested against raising strength and guidance_scale to their maxima; both made results worse, not better, ruling out a simple parameter fix). On a second failure, the pipeline falls back to the original 2D affine-warp compositor (MediaPipe keypoints + alpha-composited garment PNG on HTML5 Canvas) as the final, always-available tier. Full 3D (cloth simulation or diffusion-based mesh rendering) was considered and ruled out as out of scope for a solo build at any tier. Size-proportional rendering carries through every tier: when compositing a size other than the model-recommended one, the render is scaled to that size's measurement ratios from the size chart — an XL on a small-framed user renders visibly wider and longer than an M.

Lower-body standardization (shipped). For any top/outerwear try-on on a full-body photo, the region below the waist is always standardized to plain black fitted leggings, regardless of the shopper's actual bottoms — a consistent, catalog-style presentation independent of what is out of frame for the garment being tried on. IDM-VTON implements this as two chained calls (top first, then a second call on the first pass's own output against a fixed leggings reference), at roughly 1.7× the single-pass latency. A symmetric upper-body version — standardizing the top above a bottoms try-on — was built and tested but is **disabled in production due to a root-caused issue**: two different reference garments and both chained-call orderings each surface an IDM-VTON boundary-bleed artifact between the two chained passes. Rather than ship visually incorrect output, this feature is

documented as future work pending a fix to the chaining logic (see docs/genai_usage.md).

Size recommendation UX

Measurements are always manually entered (height, optional weight, bust via band+cup or converted chest measurement flagged lower-precision, hip, body type). The uploaded photo is for try-on pose only and never feeds the size model. The recommendation arrives in a confidence box: blue with a checkmark for confident matches; amber with a lightbulb “Sizing tip:” for borderline cases (near a size boundary + low-stretch fabric + fitted cut → recommend one size up, lower the displayed confidence). Advice text is two paragraphs: measurement-based reasoning, then a “Note:” paragraph of plain-language visual observations from the composite.

Design lesson: correlation is not causation

The first /recommend-size design scanned candidate sizes and picked the one maximizing $P(\text{fit})$ — and testing showed it inverts: a petite profile got XL at 90% “confidence”. Cause: in the training data, size_ordered tracks the customer's own body, so the model learned the between-customer correlation (larger sizes are ordered by larger people, who disproportionately report items running small), not the within-person effect of changing size. A (petite body, size 17) row is far out of distribution, and the model's counterfactual answer there is meaningless. The fix: anchor the size deterministically from measurements via the size chart, ask the model the in-distribution question (“does this size run small/fit/large for this body?” — literally what each training row records), shift one size if the top class is a misfit, then apply the borderline advisory layer. Verified sane: petite → XS, average → M, large-bodied → XL.

History-based suggestions (rule-based, not a second model)

Each try-on stores size shown, brand, fabric, and the user's fit feedback. When 2+ prior items from the same brand or fabric show consistent feedback, a one-line note is added to the advice prompt (“items from this brand have run true to size for you”) — a SQL lookup plus a conditional prompt line, deliberately not another trained model.

Privacy (full framing in docs/privacy.md)

- Uploads live under anonymous UUID session folders (backend/uploads/{session_id}/), never name/email-keyed, gitignored, session-scoped.
- 24-hour auto-deletion enforced in code (purge on startup, before every upload, and as a scheduled script); the deployed host's ephemeral disk makes redeploys a stricter version of the same guarantee.
- Crop-at-upload, opt-in (unchecked by default): a checkbox on the upload form removes everything above the nose — the upper face (forehead, eyes) — before the photo is ever written to disk, pose-extracted, or sent to any try-on engine; the lower face (mouth, chin, jaw) remains visible on this path. Left unchecked (the default), the photo is stored and processed with the face visible, and gets a defense-in-depth protection instead: a face bounding box is detected once and carried in pose.json so every generated try-on render gets the real face region re-pasted over whatever the engine drew there, with a feathered edge — closing a benchmarked IDM-VTON tendency to occasionally regenerate a synthetic, unrelated face on full-body renders.
- Third-party processing disclosure: generative try-on sends the shopper's photo (already cropped, if that checkbox was used) to Replicate to render IDM-VTON/SDXL; Replicate never receives body measurements, name, or any other profile field.
- HTTPS-only transmission; consent notice at the point of collection stating what is collected, why, and the retention period.

- Framed against the Philippines' Data Privacy Act of 2012 (RA 10173): transparency, legitimate purpose, proportionality, retention limitation, and security — with consent for the try-on feature never reused for research or auditing.

7 Generative AI Usage (Step 9, optional)

FitML uses generative / deep-learning AI strictly outside the graded ML pipeline, in three places (full write-up in docs/genai_usage.md):

Claude API — multimodal fit advice

After the try-on composite renders, the /advice endpoint sends Claude Vision the model's pre-computed size recommendation, its confidence level, and the composited try-on image — never raw measurements. Measurements are always manually entered by the shopper; Claude cannot estimate or modify them. Claude returns two paragraphs of plain-language advice that can reference visual details in the composite (how the garment sits, whether it looks loose or snug). Hard boundaries: Claude receives the model's recommendation as read-only context and never overrides it, and its visual observations are qualitative commentary only. The prompt bans tailoring jargon so the output stays readable for non-technical shoppers.

Generative virtual try-on — Claude Vision + IDM-VTON / SDXL

Claude Vision analyzes the shopper's uploaded photo and the target catalog garment to generate a structured JSON prompt (pose, framing, garment description) — never measurements. That prompt drives IDM-VTON (garment-conditioned diffusion via Replicate, the default engine), which renders the composite showing the shopper wearing the catalog item; SDXL inpainting is the fallback if IDM-VTON fails, and the original 2D affine-warp compositor is the final, always-available tier (Section 6.2 has the full three-tier detail). The analyzer prompt is hard-banned from mentioning body measurements, enforced by an automated smoke-test assertion, and no data from this pipeline feeds the graded size model — it renders the recommended size, never estimates it. Seeds default to 42 and Replicate model versions are pinned in code for reproducibility, the same discipline the graded pipeline uses.

rembg / U²-Net — catalog background removal

The catalog pipeline runs product photos through rembg's clothing-segmentation model (U²-Net, a pretrained CNN) to produce the transparent garment PNGs the compositor warps. It runs offline, once per image, sees no user data, and touches no training data.

Why GenAI is fenced off from the graded pipeline

- **Auditability:** the fairness audit makes quantitative claims about a model whose inputs and parameters are fixed and inspectable; a generative model in the prediction path would make those claims unverifiable.
- **No silent data invention:** letting a vision model estimate measurements from photos — whether for advice text or for try-on rendering — would inject unquantifiable, likely body-type-correlated error into exactly the variables the audit stratifies on.
- **Reproducibility:** every graded artifact reproduces from fixed seeds and committed code; generative output is confined to surfaces where variation is harmless (advice prose, one-time asset prep, and seeded try-on rendering).

The result: deterministic, audited ML decides the size; generative AI only renders and explains it, and helps build the demo catalog.

8 Presentations (Step 6)

Two decks accompany this report, both following the program branding (white background, AIM logo on every slide):

- **Technical deck** (Debie_Galleros_Pillar5_CapstoneProject_TechnicalDeck.pptx, audience: peers and faculty): problem framing and task type, dataset overview, EDA highlights, the four-model comparison with macro-F1 reasoning, fairness-audit methodology with before/after mitigation numbers, SHAP explainability, the data → model → Flask → website architecture, GenAI usage boundaries, and limitations + future work.
- **Business deck** (Debie_Galleros_Pillar5_CapstoneProject_BusinessDeck.pptx, audience: executives, no jargon): the returns problem (\$100B+, 35% online return rate), what FitML does in shopper language, the fairness differentiator, live-site screenshots, business impact and ROI logic, privacy and risk handling, rollout strategy, and future work.

9 GitHub Repository (Step 7)

<https://github.com/debiegalleros/fitml-capstone> — public, with the rubric-required structure (src/, notebooks/, data/, models/, backend/, frontend/, docs/, README.md, requirements.txt) and a clean, incremental commit history: one commit per meaningful unit of work with descriptive messages, phase by phase from scaffolding through deployment.

- **Reproducibility:** fixed seed 42 throughout; training config, label encoder, and pinned test indices committed alongside the model artifacts; the fairness audit refuses to run if the re-derived split stops matching the pinned indices.
- **Hygiene:** raw datasets, catalog image assets, uploaded photos, and API keys are gitignored; requirements.txt is frozen to the exact working versions; the one oversized artifact (random_forest.joblib, 50 MB) is gitignored and byte-reproducible from the fixed-seed training script.
- **Documentation:** eight docs/ files cover the data dictionary, model selection, the full fairness report, data provenance, privacy, GenAI usage, and deployment — all cross-linked from the README.

10 Limitations & Future Work

- Class imbalance remains the dominant modeling problem. Next steps beyond the single audited mitigation: probability calibration with per-group decision thresholds, focal loss, or resampling.
- Customer-grouped splitting: re-derive customer IDs upstream so repeat customers cannot straddle the train/test split.
- Fit-aware generative try-on: this submission already ships generative try-on (IDM-VTON primary, SDXL fallback) with size-proportional rendering carried through every tier; the FIT dataset (2026; 1.13M fit-aware try-on triplets) is the cited research path beyond it — a model trained specifically to render how a given size actually drapes and fits, rather than garment-conditioned compositing at a scaled size.
- SDXL's fork-specific weak prompt adherence (Section 7) is a documented, root-caused limitation of the fallback engine, not the primary one; a same-tier alternative inpainting backend is future work if IDM-VTON's Replicate availability ever becomes a concern.

- Symmetric upper-body standardization (mirroring the shipped lower-body leggings feature for bottoms try-on) was built and tested but stays disabled: two reference garments and both chained-call orderings each surfaced a genuine IDM-VTON boundary-bleed problem between the two chained passes.
- Transformer-based clothing parsing: a SegFormer-B2-clothes benchmark against the current U²-Net pipeline found no net quality advantage today (59 vs 56 clean items, 5 outright failures), but the approach is the natural upgrade as models mature.
- Color-suitability matching (does this shade complement the shopper) — identified from the start as a natural extension, deliberately out of scope.
- A CNN fit-quality assessor over the composited try-on image was considered and scoped out: no labeled dataset exists, and building one from the compositing pipeline was too time/risk-heavy for the deadline.
- Production hardening: pin CORS to the exact frontend origin, add per-group threshold monitoring, and collect the bust_input_method flag long enough to audit input-method bias with real usage data.

References & data sources

- Kaggle: Clothing Fit Dataset for Size Recommendation (ModCloth + RentTheRunway) — model training and fairness audit.
- Kaggle: Fashion Product Images Dataset (paramaggarwal, MIT license) — catalog imagery.
- Uniqlo published men's size charts (transcribed 2026-07-06) — men's extension label source.
- Republic Act No. 10173 (Data Privacy Act of 2012, Philippines) — privacy framing.
- FIT dataset (2026): 1.13M fit-aware try-on triplets — future-work research direction.
- Project repository: <https://github.com/debiegalleros/fitml-capstone> · Live demo: <https://fit-ml.netlify.app>



Part II

Technical Presentation Deck

Audience: peers and faculty — models, fairness audit, architecture



FitML: A Fairness-Audited Machine Learning System for Sizing and Intelligent Virtual Try-On in E-Commerce

Pillar 5: Capstone Project

Postgraduate Diploma in Artificial Intelligence and Machine Learning

Debie Galleros · July 2026

Technical presentation — fit-ml.netlify.app · github.com/debiegalleros/fitml-capstone



Problem framing and task type

35%

online clothing return rate (vs ~8% in physical stores)

\$100B+

annual cost of returns to retailers

2 questions

“will it fit?” and “how will it look on me?” — both drive hesitation

The data science task

- **Supervised multi-class classification:** predict reported fit outcome — fit / runs small / runs large — from body measurements + ordered size
- **Labels are real customer fit feedback** (not collaborative filtering)
- **Selection metric: macro-F1** — accuracy is blind to minority classes under a 73%-majority target

What makes it different

- **A published fairness audit** — disparate impact, equalized odds, SHAP, and a mitigation with honest before/after numbers
- Commercial sizing tools are black boxes; **FitML treats the audit as a first-class deliverable**
- **Generative AI try-on (IDM-VTON/SDXL via Replicate) gives the visual answer, with a 2D compositor as automatic fallback — not 3D**



Dataset: 251,047 real fit reviews

Kaggle “Clothing Fit Dataset for Size Recommendation” — ModCloth + RentTheRunway

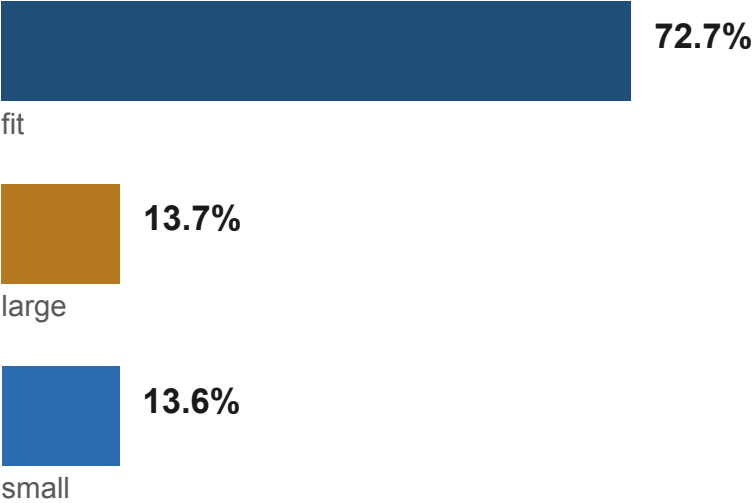
	Women's core (graded ML)	Men's extension (demo)
Customers	251,047 real reviews	200 synthetic
Labels	Real fit feedback	Uniqlo chart lookup
Sizing logic	4 trained classifiers	Deterministic rule
Fairness audit	Included (full)	Excluded (documented)

Why: both platforms sell women's apparel, so real data is women's-only. Real, citable data where it exists; a small, clearly-flagged synthetic extension for the gap — an explicit limitation, not a silently patched one.

Features (all self-reported / user-entered)

height · weight · bust band + cup · hip · body type · garment category · ordered size — target: fit / small / large

Class balance (the core difficulty)



72.7 / 13.7 / 13.6 — a model that always says “fit” gets 72.7% accuracy while being useless.



EDA and feature engineering

notebooks/02_eda_feature_engineering.ipynb · reusable logic in src/prepare_data.py

Two schemas, harmonized

Height parsed from “5ft 6in” vs “5' 8””; 68 RentTheRunway category tags mapped to a 4-bucket taxonomy; cup letters unified to an ordinal scale (AA=0 ... K=11). 24,287 ModCloth rows with no garment signal dropped.

Missingness is structural — flagged, not hidden

Weight absent for ALL ModCloth rows; hips for ALL RentTheRunway rows. Each gets a `_missing` indicator beside median imputation; `body_type` gets an explicit `not_reported` category (never a guessed shape). SHAP later proves the model uses these flags.

PCA: a useful negative result

Body-type clusters overlap heavily in 2D — components track overall size, not shape. Raw measurements won't trivially separate body types, so the fairness audit needs per-group metrics.

Leakage discipline

`model_ready.csv` ships unscaled and un-encoded; scaling and one-hot encoding are fitted inside each model's train-only Pipeline. Post-purchase fields (rating, review text) excluded as features.

Model comparison: four classifiers, one honest headline

Same stratified 80/20 split (200,837 / 50,210 rows, seed 42) · same features · preprocessing fitted train-only

model	accuracy	macro-F1	F1 fit	F1 large	F1 small
XGBoost (selected)	0.7272	0.2918	0.8418	0.0219	0.0115
Random Forest	0.7278	0.2848	0.8424	0.0101	0.0018
MLP (64/32)	0.7272	0.2825	0.8420	0.0043	0.0012
Logistic Regression	0.7267	0.2822	0.8418	0.0000	0.0046

Every unweighted model collapses toward the majority class — accuracy is pinned at the 72.7% baseline for all four; logistic regression never predicts “large” at all. XGBoost wins as the only model with non-trivial F1 on both minority classes.

Trees beat the MLP, as predicted — gradient-boosted trees dominate MLPs on modest tabular data. A legitimate finding, reported honestly (equal-footing, untuned hyperparameters for all four).

Why macro-F1?

“Small” and “large” are the whole point: they are the cases where a recommendation prevents a return.

Macro-F1 weights each class equally, exposing the minority-class failure that accuracy hides.

Accuracy would call a coin-flip between these models; macro-F1 separates them.



Fairness audit: methodology

src/fairness_audit.py re-derives the exact Phase 5 split and asserts it matches the pinned test indices

Two harms audited

- **Quality-of-service:** misfit detection failing more for some groups → wrong “it fits” reassurance → returns
- **Allocation-style:** the favorable “fit” prediction handed out at unequal rates

Groups (each $\geq 1,000$ test rows)

- **body_type** — 8 groups incl. not_reported (n=14,800)
- **height_band** — under 160 / 160–169 / 170+ cm
- **size_band** — US 0–6 / 8–14 / 16–22 / 24+

Metrics

Per-group accuracy and per-class recall · disparate impact ratio (four-fifths rule, <0.8 flagged) · equalized-odds TPR/FPR spreads

Baseline finding: “fair” numbers that mean nothing

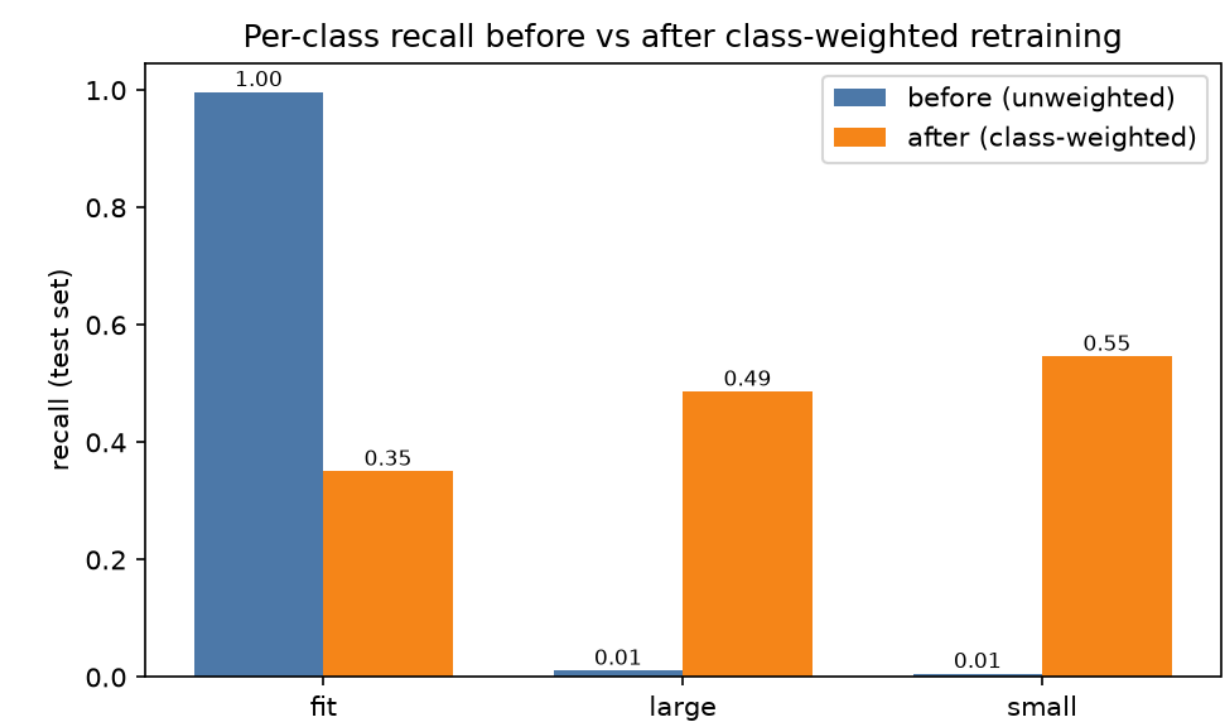
No group is DI-flagged (all ≥ 0.98) — because the model predicts “fit” for 99.5% of rows in every group. A model that always says yes passes the four-fifths rule by construction.

Inside the predictions: a 7.3-point accuracy gap against size 24+ shoppers; “runs large” caught 8.2% of the time for full-bust vs 0.34% for pear (24×); zero “small” recall for apple body types.

The fit-class FPR is 0.96–1.00 everywhere: when an item does not fit, virtually every shopper is told it does.

Mitigation: class-weighted retraining — before / after

Identical pipeline, split, and seed; the only change is balanced sample weights



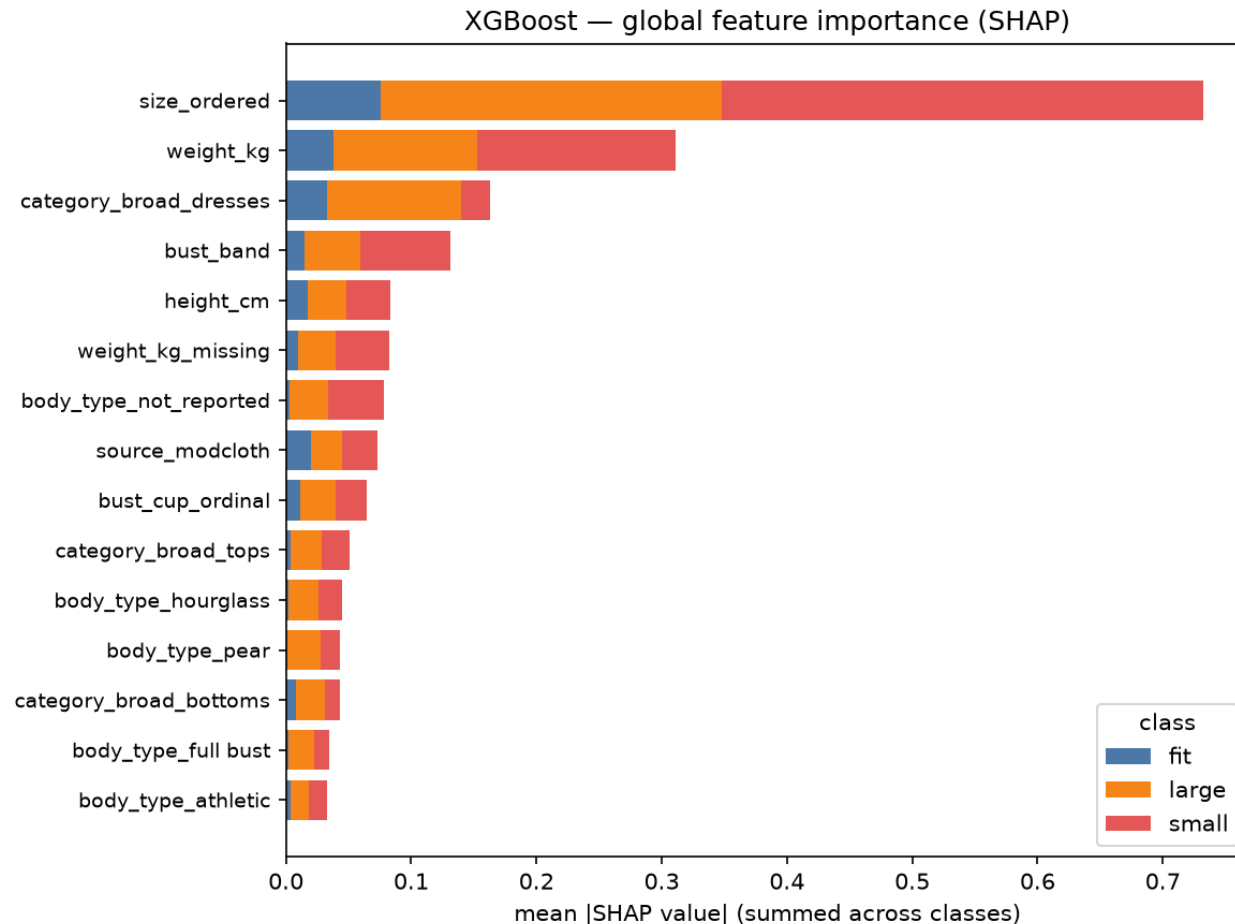
metric	before	after
macro-F1 (selection metric)	0.2918	0.3596
recall — large	0.011	0.486
recall — small	0.006	0.546
worst-group fit FPR	0.996	0.346
accuracy	0.727	0.396
DI flags	0 (vacuous)	7 (informative)

The flags after mitigation are informative, not a regression — once the model actually discriminates between classes, it surfaces the real disparity the degenerate baseline had hidden (size 24+ DI 0.41; apple/full-bust 0.64–0.70). Height stays equal (≥ 0.99).

Deployed: the class-weighted model. Catching misfits matters more than raw accuracy; the confidence % and blue/amber advisory layer absorb the uncertainty.

Explainability: SHAP on the winning model

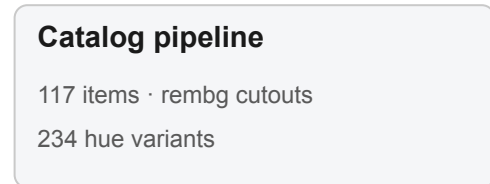
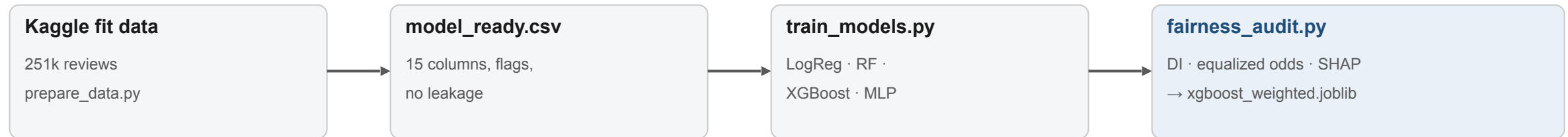
TreeExplainer · 3,000 test rows · seed 42



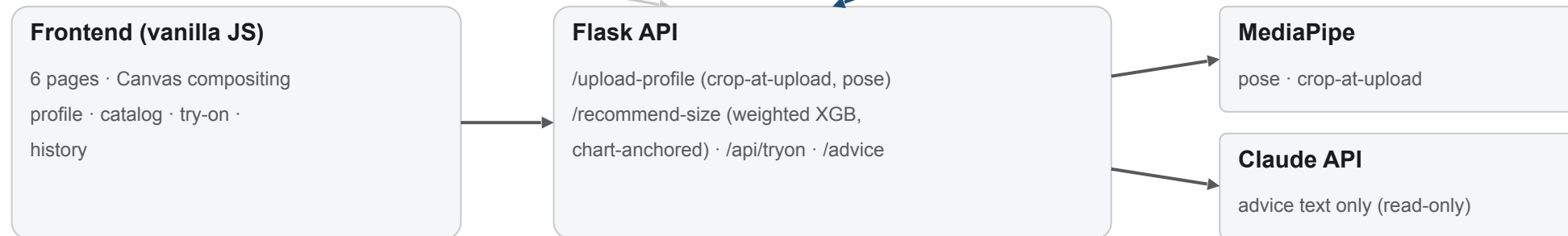
- **size_ordered dominates** (mean |SHAP| 0.38 for small, 0.27 for large vs 0.08 for fit) — and directionally correctly: high ordered size, low weight, small bust band all push toward “runs small”.
- **The model learned real sizing signal**; it is the decision threshold, drowned by class imbalance, that fails — exactly what class weighting attacks.
- **Red flag: missingness is predictive.** weight_kg_missing and body_type_not_reported rank in the top 7 for minority classes. Shoppers who disclose less get systematically different predictions — a data-completeness bias, answered with transparency in the product (confidence drops when key fields are absent).

Architecture: data to deployed product

OFFLINE — reproducible, seed 42



LIVE — fit-ml.onrender.com (Docker) + fit-ml.netlify.app



Generative AI: used deliberately, fenced off from the graded ML

Claude Vision + Replicate — generative try-on & fit advice

Claude Vision analyzes the shopper's photo + garment, producing a structured prompt; IDM-VTON (default) or SDXL inpainting renders the try-on via Replicate. Claude then reviews the composited image for two paragraphs of plain-language advice (second must start "Note:."; tailoring jargon banned).

Try-on renders the recommended size, never decides it. Advice never estimates measurements or overrides the model — a read-only consumer of the prediction.

rembg / U²-Net — catalog background removal

A pretrained segmentation CNN produces the transparent garment PNGs the try-on compositor warps.

Runs offline, once per image, at catalog build time. Sees no user data, produces no labels, touches no training data.

Why the fence?

- **Auditability** — the audit's quantitative claims need a fixed, inspectable model; a generative model in the prediction path would make them unverifiable
- **No silent data invention** — vision-estimated measurements would inject body-type-correlated error into exactly the variables the audit stratifies on
- **Reproducibility** — graded artifacts rebuild from fixed seeds; generative output is confined to advice prose and one-time asset prep



Limitations and future work

Known limitations (documented, not hidden)

- **Severe class imbalance remains the dominant problem** — one mitigation trialed, fully audited
- Random split does not group repeat customers (IDs dropped upstream)
- Self-reported measurements — noisy, plausibly in group-correlated ways
- Groups are proxies: no age/ethnicity/disability fields in the data
- Men's extension is chart-based and unaudited, by design

Future work

- **Probability calibration + per-group thresholds** to equalize misfit-detection rates — the clear next mitigation
- **FIT dataset (2026, 1.13M fit-aware try-on triplets)** — the research path beyond today's garment-conditioned rendering at a scaled size, toward true per-size drape
- **Transformer clothing parsing** (SegFormer benchmarked: no net win today — 59 vs 56 clean items, 5 failures)
- **Color-suitability matching** — scoped out from the start
- **bust_input_method audit** once real usage data exists



Part III

Business Presentation Deck

Audience: executives — the returns problem, the product, impact and rollout



FitML: A Fairness-Audited Machine Learning System for Sizing and Intelligent Virtual Try-On in E-Commerce

Pillar 5: Capstone Project

Postgraduate Diploma in Artificial Intelligence and Machine Learning

Debie Galleros · July 2026

Business presentation — live demo: fit-ml.netlify.app

Returns are eating online fashion's margins

35%

of online clothing orders come back — vs about 8% in physical stores

\$100B+

what returns cost retailers every year

#1

cause: sizing uncertainty — shoppers can't try before they buy

Two questions stop a shopper from clicking “buy”:

“Will it fit?”

Every wrong answer is a return — shipped twice, repacked, often discounted.

“How will it look on me?”

If the shopper can't picture it, they don't buy — or they order three sizes to try at home.

FitML answers both — before the purchase

A working prototype, live at fit-ml.netlify.app

1

Set up a profile once

The shopper enters their measurements and uploads one photo. Nothing is guessed from the photo — it is only used to show garments on them.

2

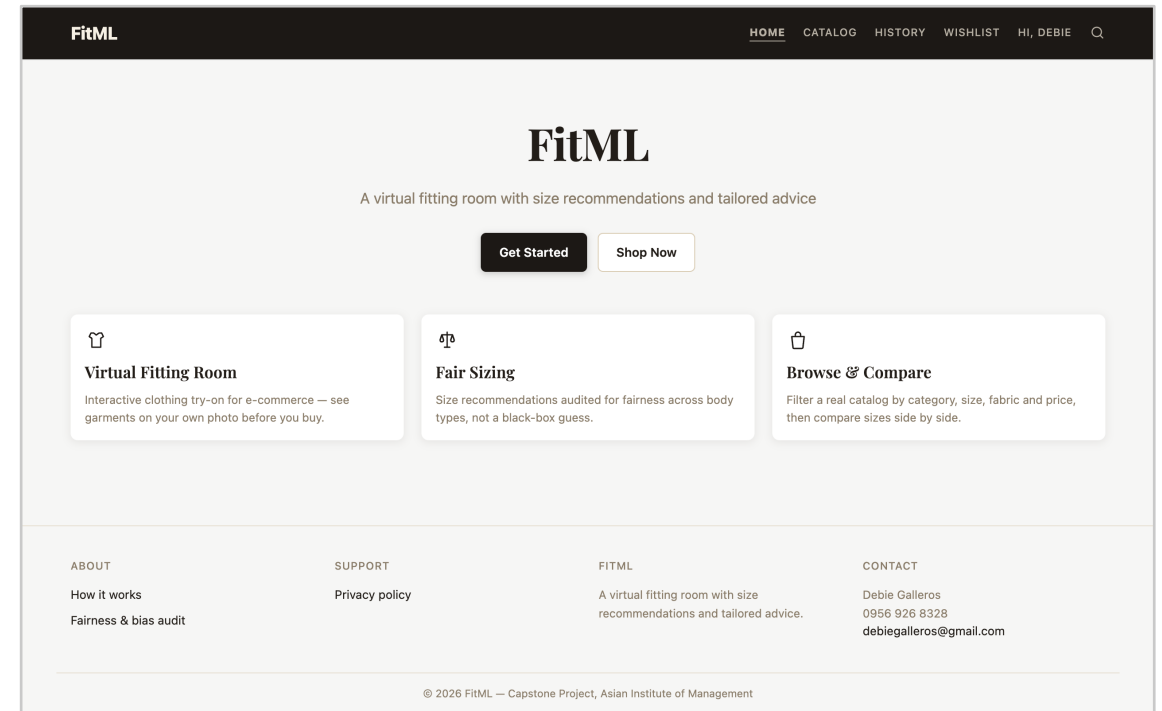
Browse and try on

117 real items with filters, prices, and color options. One tap shows the garment on the shopper's own photo.

3

Get a size and honest advice

A recommended size with a confidence level, plus two short paragraphs of plain-language advice — including what the try-on photo actually shows.



The live site

The virtual fitting room, on the shopper's own photo



Live try-on: recommendation, composite, and advice for a real catalog item

A clear recommendation

“Recommended size: M — 45% confidence” — the confidence level is shown, never hidden.

The garment on their body

The blouse is placed on the shopper's own photo, scaled and angled to their pose — picking a bigger size renders visibly bigger.

Advice a shopper actually understands

Plain words, no tailor talk: “the short sleeves sit neatly on your arms, and the buttons run straight and even down the front.”

Privacy by default

The upper face (forehead, eyes) can be cropped out before the photo is ever saved, opt-in — shown here live on the current site.



The differentiator: sizing that is checked for everyone

Most sizing tools are black boxes. FitML publishes its audit.

Sizing models are usually most accurate for “average” bodies — and quietly worse for everyone else. We measured ours across body types, heights, and size ranges, found exactly that problem, and retrained to fix what matters most.

What we found: the first model told almost every shopper “it fits” — it was most wrong for plus-size shoppers, exactly the people returns hit hardest.

What we did: retrained the model to take misfits seriously and published before/after numbers for every group — the full report is in the open repository.

1 in 100 → 1 in 2

misfit warnings the model catches, before vs after the fairness fix — the single number behind fewer wrong “it fits” promises

8 groups × 3 lenses

body types, heights, and size bands — every group's numbers published, none averaged away

0 guesses

measurements are always entered by the shopper — never estimated from a photo



What this is worth

Projected impact — to be validated in a pilot, not promised in advance

The mechanism

- **Fewer wrong “it fits” moments.** The retrained model catches about half of misfits before purchase, instead of almost none.
- **Fewer “order 3, keep 1” baskets.** Seeing the garment on their own photo replaces ordering multiple sizes to try at home.
- **Higher purchase confidence.** A visible confidence level and honest sizing tips build trust — and trust converts.

Every avoided return saves twice-paid shipping, inspection, repackaging, and markdown loss — and keeps the item sellable at full price.

The target

35% → 15–20%

return-rate goal for sizing-driven returns, in line with what published fit-technology deployments report

Illustrative: a shop doing 10,000 orders a month at 35% returns processes 3,500 returns. Cutting the rate to 20% removes 1,500 of them — every month, before counting the sales that no longer get lost to hesitation.



Risks handled up front

Body photos are sensitive — the prototype treats them that way, by design



Photos deleted in 24 hours

Auto-deletion is enforced in code, not policy. Uploads live in anonymous, session-scoped folders — never linked to a name or email.



Face privacy, opt-in and automatic

An opt-in checkbox crops the upper face — forehead and eyes — out before saving. Left unchecked, FitML still protects you — your real face is restored over anything the AI drew in its place.



No body measurements are ever guessed

The photo is used only to show garments. Measurements come from the shopper, full stop — no silent AI estimation of anyone's body.



Compliant framing from day one

Built against the Philippines' Data Privacy Act (RA 10173): consent up front, one declared purpose, minimal collection, hard retention limits.

And the honest risk register

- The model over-warns today: catching misfits costs some false sizing tips. The interface presents them as tips, not verdicts.
- Trained on US women's rental/retail data; local fit habits will differ. The pilot exists to recalibrate on real local outcomes.
- Men's sizing is a size-chart lookup, clearly labeled — not the audited model.

Rollout: pilot, measure, then scale

Phase 1 — Pilot

8–10 weeks

One women's category (dresses) on the existing store. Measure return rate, size-exchange rate, and conversion against a control group.

Phase 2 — Calibrate

next quarter

Retune the model's thresholds on real local outcomes, per body-type group — the audit framework is already built for exactly this.

Phase 3 — Scale

quarter after

Full women's catalog, then men's once enough real fit feedback exists to train and audit a proper men's model.

The gate between phases is a number, not a feeling: the pilot ships with the same measurement dashboard the capstone already produces — per-group accuracy, misfit-catch rate, and return-rate delta.



Where this goes next

Fit-accurate rendering

Today's generative try-on already shows the right garment on your own photo, photorealistically. The next step (the FIT dataset, 2026 — over a million fit-aware examples) renders how a specific size actually drapes and pulls, not just where it sits.

Smarter sizing over time

Every confirmed purchase and return becomes training signal. The fairness dashboard keeps the improvement honest across body types.

Color advice

“Does this shade suit me?” — scoped out of the capstone deliberately, and a natural add-on to the same advice panel.

Richer men's experience

Real men's fit data from the pilot replaces today's clearly-labeled size-chart lookup with an audited model.



Try it yourself

fit-ml.netlify.app

The full fairness audit, code, and documentation are public: github.com/debiegalleros/fitml-capstone

Debie Galleros · Postgraduate Diploma in Artificial Intelligence and Machine Learning · July 2026